Optimal Laser-Beam Engagement of Fast-Moving Inertial Fusion Energy Targets

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Pierre Boca
pboca@stanford.edu

1 Project Status & Revisions

The baseline Kalman filter (KF) formulation — state estimation feeding a strictly convex least-squares optimizer extended via Sample Average Approximation (SAA) for non-Gaussian noise — is pivoting toward a robust, data-driven formulation that addresses two real-world complexities raised from a hardware perspective: (1) calibration drift in beam pointing, (2) spatially correlated chamber noise. These extensions map naturally to recent course topics: **Probabilistic Surrogate Models**, **Optimization under Uncertainty (OUU)**.

2 Revised Methodology

2.1 Probabilistic Surrogate Modeling for Chamber Dynamics

The original model assumed i.i.d. process noise $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$. However, small-scale correlated gas fluctuations produce structured, spatially correlated perturbations. We replace the fixed $\mathbf{Q}$ with a **Gaussian Process** surrogate trained on the process residuals of the I preceding target trajectories:

$$\boldsymbol{\delta}_k^{(n)} = \mathbf{x}_k^{(n)} - \mathbf{F}\mathbf{x}_{k-1}^{(n)}, \quad n \in \{n-I, \dots, n-1\},$$

modeled as $\boldsymbol{\delta}(\mathbf{r}) \sim \mathcal{GP}(\mu_{\mathrm{GP}}(\mathbf{r}),\, k_{\mathrm{SE}}(\mathbf{r}, \mathbf{r}'; \sigma_f, \ell))$ with a squared-exponential kernel whose hyperparameters (σ_f, ℓ) are fitted by marginal-likelihood maximization. The GP posterior mean and covariance at the predicted target location replace the isotropic $\mathbf{Q}$ in the KF predict step, yielding a data-informed noise model that sharpens the position posterior $p(\mathbf{r}(t_f) \mid \mathbf{z}_{1:m})$ when chamber conditions are consistent across shots.

2.2 Uncertainty Propagation & Calibration Drift

Calibration drift introduces a slowly growing offset between the commanded focal point and its true position. We model the nominal focus as $\mathbf{o}_i = \bar{\mathbf{o}}_i + \Delta\mathbf{o}_i$, where $\Delta\mathbf{o}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{cal}})$ is an *epistemic* uncertainty whose covariance grows with time since last calibration. Propagating this through the focal-point model $\mathbf{p}_i(\mathbf{u}_i) = \mathbf{L}_i\mathbf{u}_i + \mathbf{o}_i$ yields a **robust Optimization under Uncertainty objective**:

$$\min_{\mathbf{u}} \max_{\Delta\mathbf{o} \in \mathcal{U}_\epsilon} \mathbb{E}\left[\sum_{i=1}^{N} \left\|\mathbf{L}_i\mathbf{u}_i + \bar{\mathbf{o}}_i + \Delta\mathbf{o}_i - \hat{\mathbf{r}}(t_f)\right\|^2 \,\middle|\, \mathbf{z}_{1:m}\right], \tag{1}$$

where $\mathcal{U}_\epsilon = \left\{\Delta\mathbf{o} : \Delta\mathbf{o}^\top \boldsymbol{\Sigma}_{\mathrm{cal}}^{-1} \Delta\mathbf{o} \le \chi^2_{3,\,0.95}\right\}$ is the 95th-percentile ellipsoidal uncertainty set. Rather than minimizing expected error alone, this formulation hedges against the worst-case drift realization. The inner maximization admits a closed form via Cauchy–Schwarz, so the overall problem remains a quadratic program solvable by the existing log-barrier interior-point method.

3 Updated Measures of Success

1. **Surrogate accuracy:** GP leave-one-out cross-validation RMSE on held-out shots, compared against the i.i.d. $\mathbf{Q}$ baseline.
2. **Robust optimizer:** $\varepsilon_{\mathrm{rms}}$ of the OUU solution vs. nominal (expected-value) optimizer as drift magnitude $\|\Delta\mathbf{o}\|$ is swept from 0 to 50 μm.

4 Time line and next steps

1. **Feedback from the industry on my model:** A meeting with **Inertia** a start-up in the Inertial Fusion Energy sector – in which I am interning this summer – is scheduled to discuss the model and validate the physical assumptions behind fast-moving target tracking and laser-beam engagement.
2. **Implementation:** The work will now focus on implementing the optimization problem to validate the measures of success.